

AADK2 – TASK 2

Student Name: Rajat Prakash Dhal
Course/Unit: Android Development with Kotlin
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Project Name: GreetingCard_RajatPrakashDhal

What Are Composables?

In Jetpack Compose (used in **Android Studio**), UI elements are built using **@Composable functions**. These functions describe how the UI should look and behave. Instead of XML layouts, Compose uses layout primitives like **Column, Row, and Box** to structure the screen.

These primitives are the foundation of modern Android UI design.

Column

◆ Definition

Column is a layout composable that places its children **vertically (top to bottom)**.

◆ Typical Use Cases

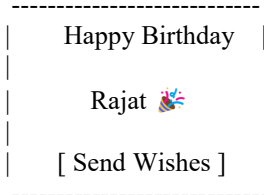
- Displaying a vertical list of items (Title → Subtitle → Button)
- Forms (Name field → Email field → Submit button)

◆ Limitation / Gotcha

- Items stack vertically by default; spacing and alignment must be managed using Arrangement and Alignment.

Example UI — Greeting Card Layout

- **Use Case:**
Stacking elements vertically.
- **Wireframe (Phone Portrait)**



- **Why Column?**

Use Column to stack **Title** → **Name** → **Button** vertically in order.

Row

- ◆ **Definition**

Row is a layout composable that places its children **horizontally (left to right)**.

- ◆ **Typical Use Cases**

- Toolbar (Back icon + Title + Menu icon)
- Buttons placed side by side

- ◆ **Limitation / Gotcha**

- Content may overflow horizontally if not managed with spacing or weight.

Example UI — Toolbar Layout



- **Why Row?**

Row is ideal for placing **Icon** → **Title** → **Action icon** in a horizontal line.

Box

- ◆ **Definition**

Box stacks children **on top of each other (overlapping layout)**.

- ◆ **Typical Use Cases**

- Image with overlay text
- Notification badge on an icon

◆ **Limitation / Gotcha**

- Children overlap by default; alignment must be controlled manually.

 Example UI — Notification Badge



- **Why Box?**

Box allows placing a **badge (3)** on top of the notification icon.

Comparison Table

Primitive	Direction	Best For	Limitation
Column	Vertical	Forms, vertical lists	Needs spacing control
Row	Horizontal	Toolbars, button groups	Risk of overflow
Box	Overlapping	Badges, image overlays	Manual alignment required

@preview

